

A universal endpoint profile for all comparison columns

Asymptotic research note; independently reviewed

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For every m , let $p_1, \dots, p_m : [K] \rightarrow [0, 1]$ be nondecreasing, with positive probability weights w_q , and suppose

$$\mathbb{E}_w \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]).$$

The number of rows and weights can depend on m . Put $A = \prod_j p_j$, $\eta(q) = (m + 1)w_q A(q)$ and, on $A > 0$,

$$r_j = \frac{1 - p_j}{p_j}, \quad \lambda = \sum_j r_j.$$

The separately proved endpoint theorem in `../independent_directions/endpoint_poisson_rigidity.tex` shows that λ under η converges to $\text{Exp}(1)$, and that $\eta(\max_j r_j \geq \varepsilon) \leq 1/(m\varepsilon)$. The present note strengthens that conclusion from the sum of the odds to every individual column, uniformly.

Lemma 1 (A fixed number of endpoint failures). *Fix an integer $\ell \geq 0$, and choose any ℓ columns $J = J_m$. Define a probability measure on the original rows by*

$$\nu_J(q) = (m + 1) \binom{m}{\ell} w_q \prod_{j \in J} (1 - p_j(q)) \prod_{i \notin J} p_i(q).$$

On its support set $\lambda_{-J} = \sum_{i \notin J} (1 - p_i)/p_i$. As $m \rightarrow \infty$, its distribution under ν_J converges to the Gamma law of shape $\ell + 1$ and rate one. The convergence is uniform in the choice of J , in the sequential sense that it holds for every sequence J_m .

Proof. A particular failure pattern of size s has probability $1/((m+1)\binom{m}{s})$, by expanding the mixed moments. Writing $d = m - \ell$, the exact generating polynomial for additional failures is consequently

$$(m+1)\binom{m}{\ell}\mathbb{E}_w\left[\prod_{j\in J}(1-p_j)\prod_{i\notin J}(p_i+z(1-p_i))\right]=\sum_{s=0}^d\binom{\ell+s}{\ell}z^s.$$

For any one index $i \notin J$, replace its factor on the left by $1-p_i$. The resulting coefficient of z^s is

$$\frac{s+1}{d}\binom{\ell+s+1}{\ell}.$$

Thus, for $0 \leq z < 1$, this polynomial is at most $(\ell+1)/(d(1-z)^{\ell+2})$.

The event that an odds ratio outside J is at least ε is an initial segment. Choosing a column witnessing this event at its last row bounds its contribution to the first generating polynomial by

$$(\varepsilon^{-1}+z)\frac{\ell+1}{d(1-z)^{\ell+2}}.$$

This is precisely the bad-prefix argument from the endpoint theorem; using the undivided polynomial includes rows containing zero entries. Choose $\varepsilon = d^{-1/2}$. On the complementary good rows, $\prod_{i\notin J}(1+zr_i)$ is squeezed between exponential moments with parameters z and $(1+\varepsilon)z$, as in that theorem. Their common limit is $(1-z)^{-\ell-1}$. The omitted mass tends to zero. Exponential tightness, uniform integrability at a smaller positive parameter, and uniqueness of moment generating functions identify the claimed Gamma limit. All bounds depend only on ℓ, d, z, ε , not on J . $\square$

Theorem 1 (Uniform profile). *For every fixed $L < \infty$,*

$$\sup_{\substack{q:A(q)>0 \\ \lambda(q)\leq L}}\max_{1\leq j\leq m}|mr_j(q)-\lambda(q)|\longrightarrow 0.$$

Equivalently, the same assertion holds with r_j replaced by $1-p_j$. It permits flat columns, unequal row weights, and different choices of the representations at every m .

Proof. Choose any sequence of indices $j = j_m$. Put $\lambda_{-j} = \lambda - r_j$ on $A > 0$. Under η , its law converges to $\text{Exp}(1)$, because $0 \leq r_j \leq \max_i r_i \rightarrow 0$ in probability. This convergence is uniform in the choice of j . By Lemma 1, under ν_j the law of λ_{-j} converges to $\text{Gamma}(2, 1)$.

We claim that $r_j \rightarrow 0$ in ν_j probability as well, assigning $r_j = +\infty$ where $p_j = 0$. Fix $\varepsilon > 0$ and let $D_j = \{r_j \geq \varepsilon\}$, an initial segment. Its η -mass is at most $1/(m\varepsilon)$. For every fixed $B < \infty$, all rows of D_j that have positive ν_j -mass must eventually satisfy $\lambda_{-j} > B$. Otherwise take the last such row in D_j with $\lambda_{-j} \leq B$. Since λ_{-j} decreases with the row,

$$\eta(\lambda_{-j} > B) \leq \eta(D_j) \rightarrow 0,$$

contradicting its exponential limit. Rows where an odds ratio outside j is infinite have zero ν_j -mass and can be omitted. Tightness of the Gamma limit therefore gives $\nu_j(D_j) \rightarrow 0$. It follows that, after discarding zero- A rows of vanishing ν_j -mass, the law of the full λ under ν_j also converges to $\text{Gamma}(2, 1)$. On $A > 0$ the exact density relation is

$$d\nu_j = mr_j d\eta.$$

Now choose any sequence of rows with $\lambda(q_m) \rightarrow x < \infty$. For an interval I lying strictly above x , monotonicity implies $mr_j(q_m) \leq \nu_j(\lambda \in I)/\eta(\lambda \in I)$ for all large m . For an interval strictly below x , the inequality is reversed. Every interval with positive length inside $(0, \infty)$ has positive limiting η -mass, and the ratio tends to

$$\frac{\int_I te^{-t} dt}{\int_I e^{-t} dt},$$

which lies between the endpoints of I . Letting the bounding intervals approach x gives $mr_j(q_m) \rightarrow x$. For $x = 0$, use only the upper interval bound and nonnegativity. If the displayed uniform assertion failed, a sequence of offending indices and rows would have a subsequence with $\lambda(q_m)$ converging in $[0, L]$, contradicting this conclusion. Finally, uniformly on that region mr_j is bounded and

$$0 \leq mr_j - m(1 - p_j) = \frac{mr_j^2}{1 + r_j} = O(1/m),$$

proving the equivalent form. □

Corollary 1 (The original tail mass is a linear coordinate). *For every fixed $L < \infty$, uniformly in $0 \leq x \leq L$,*

$$(m+1) \sum_{q: A(q) > 0, \lambda(q) \leq x} w_q \rightarrow x.$$

For any row q with $A(q) > 0$ and $\lambda(q) \leq L$, put $u_q = (m+1) \sum_{s=q}^K w_s$. Then, uniformly in these rows and columns,

$$\lambda(q) - u_q \longrightarrow 0, \quad m(1 - p_j(q)) - u_q \longrightarrow 0.$$

For equal weights, $u_q = (m+1)(K - q + 1)/K$.

Proof. Theorem 1 gives, uniformly for $\lambda \leq L$,

$$-\log A = \sum_j \log(1 + r_j) = \lambda + O\left(\left(\max_j r_j\right)\lambda\right) = \lambda + o(1).$$

Therefore the finite measures

$$(m+1) \sum_{q: \lambda(q) \leq L} w_q \delta_{\lambda(q)} = A^{-1} \mathbf{1}_{[0, L]} d\eta$$

converge weakly to Lebesgue measure on $[0, L]$, by the exponential limit and the uniform approximation $A^{-1} = e^\lambda + o(1)$. The cumulative functions converge uniformly because the limiting measure has continuous cumulative function; extending the endpoint slightly avoids any truncation ambiguity at L . At a row of value $\lambda(q)$, its suffix mass lies between the strict and non-strict cumulative masses at that value. Their difference is a jump of the cumulative function, whose maximum tends to zero. Thus $u_q - \lambda(q) \rightarrow 0$ uniformly. The column conclusion follows from Theorem 1. $\square$

These are asymptotic structural statements. The independently proved bound $K \geq cm \log m$ is stronger than the superlinear consequence of the tail profile alone. This note supplies no further growth rate for K .